

Developer Tech Stacks & Workforce Demographics

Data Analyst Capstone Project | IBM Professional Certificate

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OUTLINE

- Executive Summary
- Introduction
- Methodology
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 - Visualization – Charts
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- Discussion
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EXECUTIVE SUMMARY I



- **Production Baseline & Web Dominance**
 - **JavaScript & Web Core:** JavaScript, HTML/CSS, and SQL represent the undisputed foundation of active production systems across global engineering teams.
- **Infrastructure, Cloud & Database Modernization**
 - **Multi-Database Ecosystems:** PostgreSQL lead production relational storage, with hybrid setups (e.g., pairing PostgreSQL with **Redis** for caching or **SQLite** for local storage) representing modern standard practice.
 - **Cloud Architecture:** Infrastructure choices solidify around **AWS** and **Azure**, with modern edge platforms gaining rapid traction.
- **AI Tool Adoption & Continuity**
 - Adoption of AI development tools shows strong continuity between current usage and future demand—centered heavily on core workflow productivity like **writing code**, **searching for answers**, and **debugging**.



EXECUTIVE SUMMARY II



- **Core Mid-Career Demographic:** The developer workforce is heavily concentrated in early-to-mid career professionals, with nearly **70% of respondents aged between 25 and 44 years old** (led by the 25–34 bracket).
- **Near-Universal Formal Education:** Formal higher education remains a critical entryway into the software industry, dominated by **Bachelor's degree** overall, while **Master's degrees** become increasingly prevalent in 35+ age cohorts, reaching near **100% formal degree attainment** among senior developers (45+).



INTRODUCTION I



- **Purpose of the Report**

- Analyse global developer survey data to identify key trends in technology adoption, tools, and future technology interest.

- **Target Audience**

- **Engineering Leadership & Hiring Managers:** Optimize recruitment strategies and guide future tech stack selection.
- **Technology Strategy Teams:** Assessing market penetration of emerging AI tools, cloud platforms, and database ecosystems.
- **Software Engineers & Career Changers:** Seeking insights on high-demand skills, and career progression paths.



INTRODUCTION II

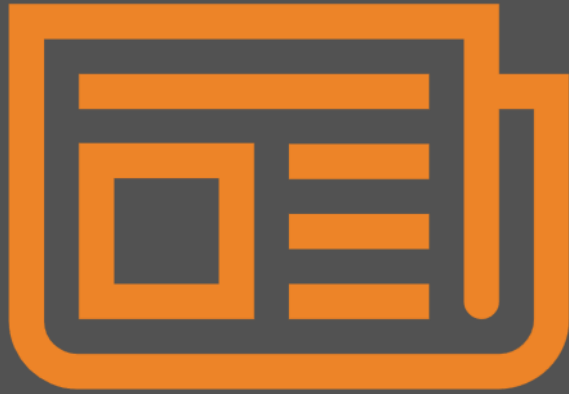


- **Business Value of the Analysis**

- **Strategic Hiring:** Data-driven insight into developer preferences (e.g., top languages, frameworks, and desired AI tools) to attract top talent.
- **Resource Allocation:** Understanding which infrastructure technologies lead in the market to prevent costly tech-stack dead ends.
- **Data-Driven Market Positioning:** Bridge the gap between active production baselines and future technology demand to make informed, high-ROI engineering investments.



METHODOLOGY I



- **Dual Data Sources**

- Real-time job postings extracted via REST API endpoints and web scraping scripts.
- Global developer insights captured from the official Stack Overflow Developer Survey dataset.

- **Automated Collection Strategy**

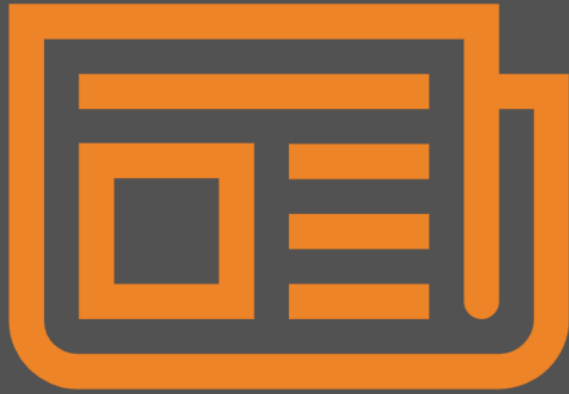
- Executed custom Python scripts using requests and BeautifulSoup to map skill distributions across major tech hubs with automated pipeline fallback logic.

- **Storage & Payload Preparation**

- Processed raw API JSON payloads into structured, multi-sheet Excel workbooks (Jobs by Location, Jobs by Technology) and standardized skill terminology for SQL analysis and Cognos dashboarding.

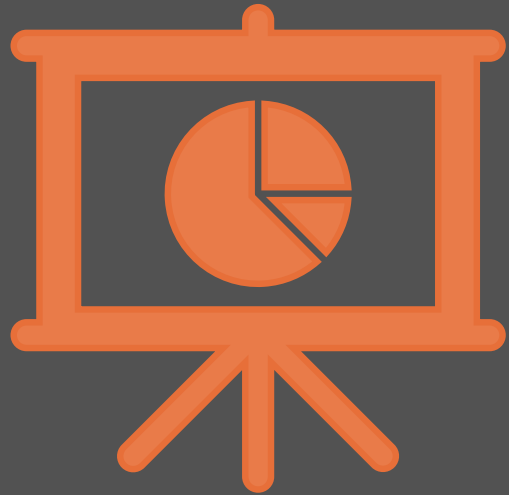


METHODOLOGY II



- **Data Cleaning & Feature Engineering**
 - **Handling Multi-Select Survey Strings:** Exploded semi-colon-delimited columns (e.g., LanguageHaveWorkedWith) via `.str.split(";")` and `.explode()` to calculate true individual technology counts.
 - **Filtering & Normalization:** Dropped non-informative NULL/NaN entries to preserve statistical integrity and built cross-tabulations across demographic brackets for 100% stacked composition charts.

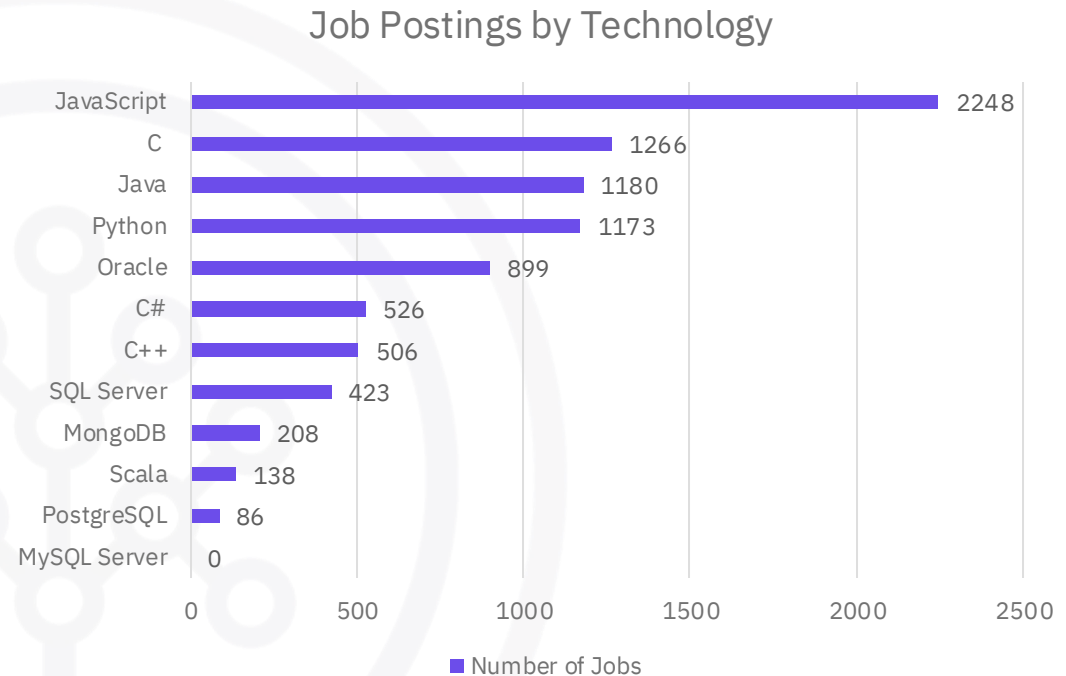
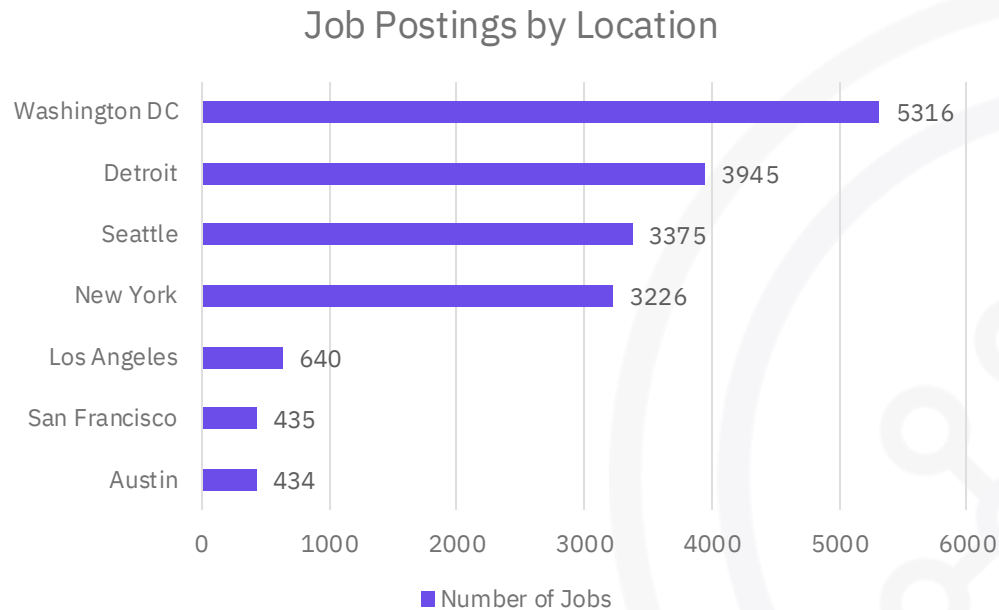




Results

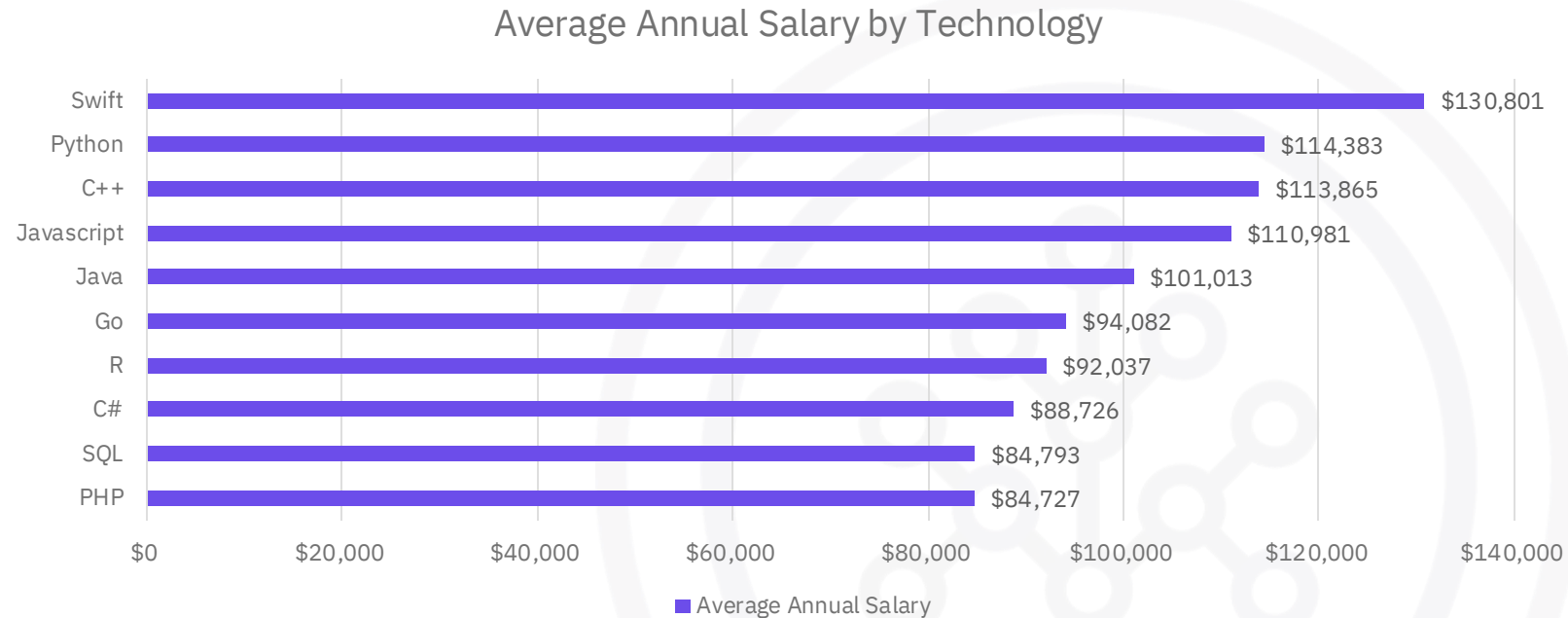
- Findings were synthesized across **Microsoft Excel** (data cleaning and preliminary metrics), **Jupyter Notebooks / Python** (scraping, EDA and visualizations), and **IBM Cognos Analytics** (interactive visual dashboards).
- Incorporated **StackOverflow** survey data with real-world job posting trends to provide a complete picture of developer skills, adoption metrics, and market demand. As well as current tool usage and developers learning preferences.
- Analyzed the global talent pool by age distribution, geographic concentration, and formal educational attainment.

JOB POSTINGS



Washington, D.C. leads all regions in total job posting volume, while **JavaScript** dominates employer demand—appearing nearly **2x as often** as the second most requested technology.

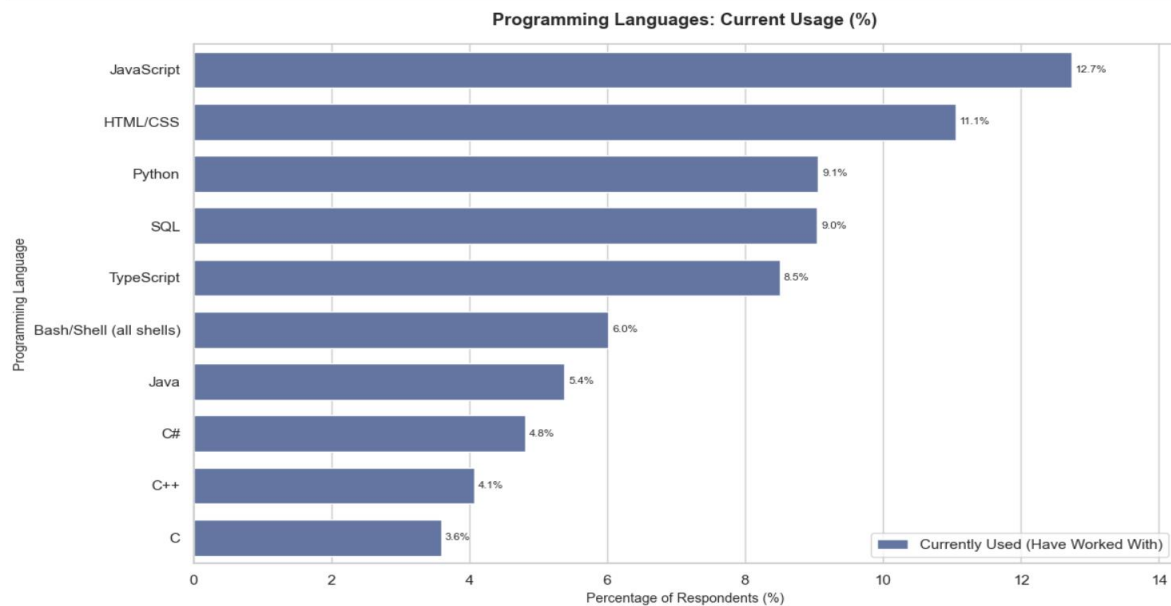
POPULAR LANGUAGES



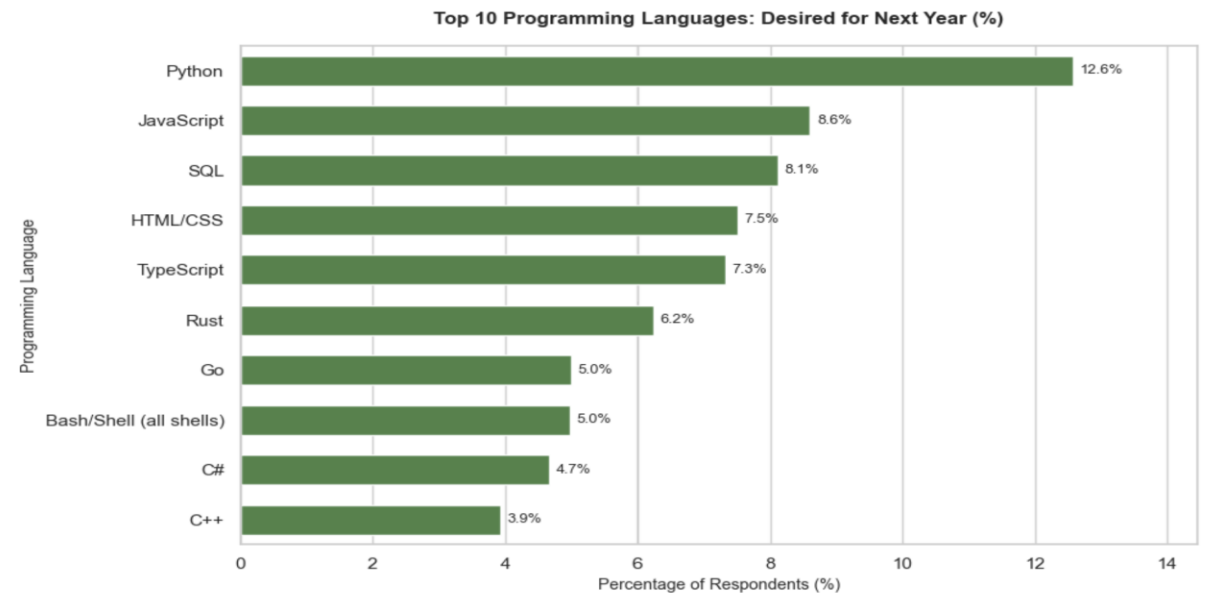
Swift commands the highest average annual salary at \$130,801, followed closely by **Python**, **C++**, and **JavaScript** clustered around ~\$110,000. **PHP** represents the lowest tier among the analyzed technologies, averaging \$84,727.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



Python takes the top spot for desired languages 12.6%, overtaking **JavaScript** 8.6% driven heavily by the rapid rise of Data Science, Machine Learning, and AI pipelines. **Rust** and **Go** climb directly into the Top 10 for next year. Traditional **C**-family and web stacks see a relative drop in desired interest compared to their current enterprise footprint.



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **Python Takes #1:** Python leads desired languages for next year at **12.6%**, overtaking JavaScript (**8.6%**).
- **Emergence of Rust & Go:** Rust (**6.2%**) and Go (**5.0%**) climb directly into the Top 10 desired languages.
- **Legacy Decline:** Traditional languages (Java, C, C++) and foundational web tools (HTML/CSS) see a relative drop in desired interest compared to their current usage footprint

Implications

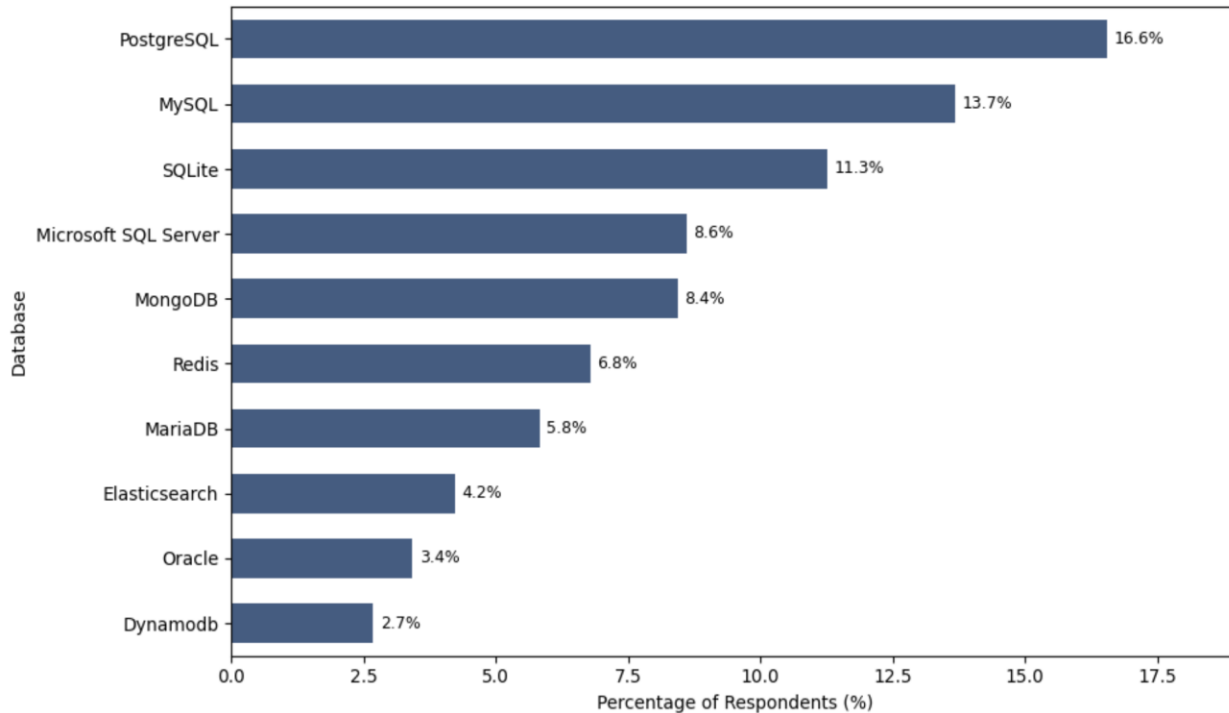
- **AI & Data Pipeline Surge:** The pivot toward Python reflects the massive global industry shift toward AI, Machine Learning, and Data Engineering ecosystems.
- **Modern Systems Engineering:** Growing demand for Rust and Go highlights a push toward high-performance, memory-safe backend services and cloud-native architecture.
- **Skills Evolution:** Developers are shifting focus from static legacy web maintenance toward specialized skills in automation, cloud computing, and high-concurrency systems.



DATABASE TRENDS

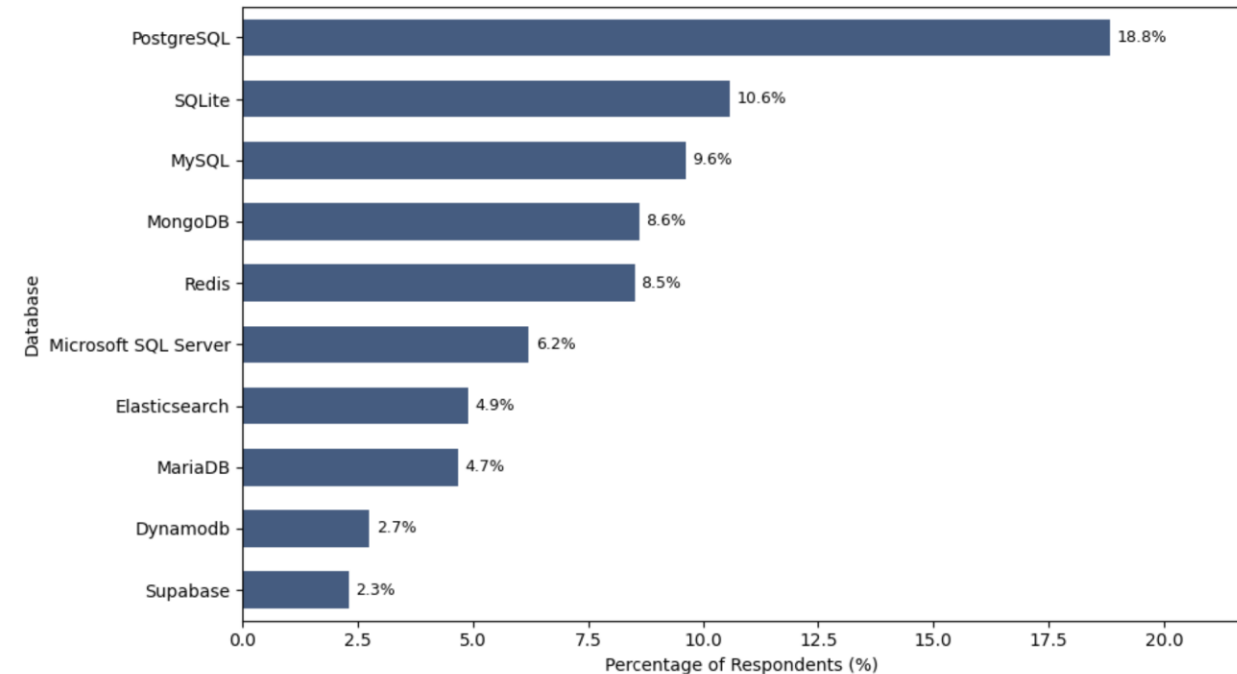
Current Year

Top 10 Databases: Current Usage (%)



Next Year

Top 10 Databases: Future Usage (%)



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **Overall High Stability:** Database preferences show much greater year-over-year stability compared to programming languages, with core platforms holding their ground.
- **PostgreSQL Solidifies #1:** PostgreSQL leads current usage at **16.6%** and extends its dominance to **18.8%** for future usage.
- **SQLite Overtakes MySQL:** SQLite jumps ahead of MySQL into 2nd place (10.6% vs 9.6%), reflecting growing demand for lightweight, embedded, and local-first data storage.
- **Oracle Exits Top 10:** Traditional enterprise legacy database **Oracle** (3.4% current) drops out of the Top 10 desired databases entirely for next year while modern cloud-managed platforms like **Supabase** (2.3%) enter the Top 10 for next year.

Implications

- **Open-Source Dominance:** Developers and organizations are increasingly consolidating around PostgreSQL as the default relational standard for both enterprise applications and modern cloud stacks.
- **Shift from Legacy On-Premise to Managed Cloud:** The exit of Oracle alongside gains for DynamoDB and Supabase highlights a clear industry transition away from high-overhead, proprietary databases toward developer-friendly, serverless, and cloud-native solutions.
- **Lightweight Architecture Preference:** SQLite's rise underscores the popularization of edge computing, local development environments, and embedded mobile/desktop storage workflows.



DASHBOARD

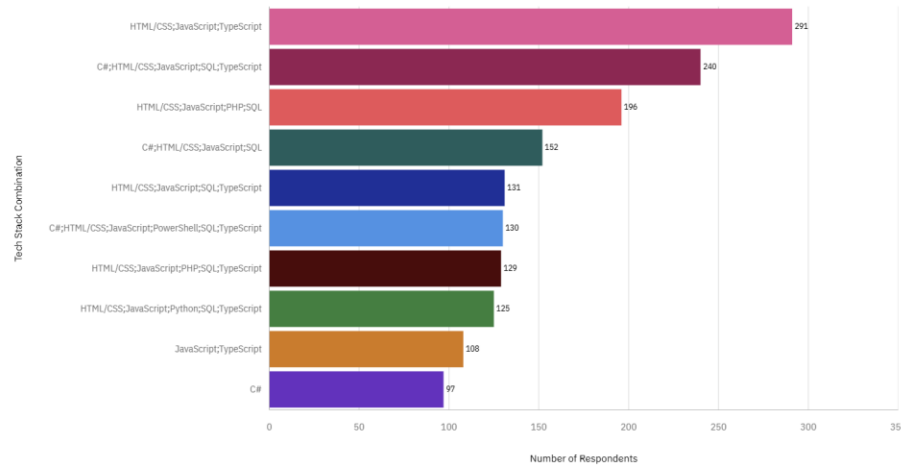


- Built using **IBM Cognos Analytics**, this interactive dashboard provides dynamic visual insights into current software adoption, future technology shifts, and global workforce demographics.
- Enables real-time exploration of active production tools alongside emerging developer learning preferences across programming languages, databases, web frameworks, and cloud platforms.
- Equips leadership and engineering stakeholders with actionable intelligence to align tech stack investments, optimize hiring strategies, and support long-term organizational expansion.

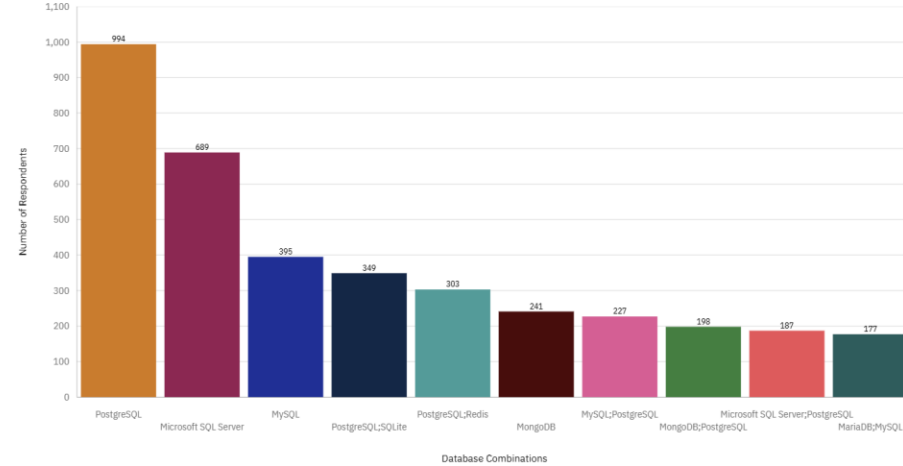


DASHBOARD TAB 1: Current Technology Usage

Top 10 Most Common Tech Stack Combination



Top 10 Most Common Databases Combinations



Top 10 Platform Combinations



Top 10 WebFrame Combinations



Microsoft Azure
Amazon Web Services (AWS)
Amazon Web Services (AWS);Microsoft Azure
Amazon Web Services (AWS);Google Cloud
Amazon Web Services (AWS);Google Cloud;Microsoft Azure
Amazon Web Services (AWS);Google Cloud;Digital Ocean
Amazon Web Services (AWS);Google Cloud;Microsoft Azure;Digital Ocean
Amazon Web Services (AWS);Google Cloud;Microsoft Azure;Digital Ocean;Digital Ocean
Amazon Web Services (AWS);Google Cloud;Microsoft Azure;Digital Ocean;Digital Ocean;Digital Ocean



DASHBOARD TAB 1: Current Technology Usage

- **Top 10 Most Common Programming Language Combinations**
 - **Findings:** The combination of **HTML/CSS, JavaScript, TypeScript** leads with **291 respondents**, followed closely by **C#, HTML/CSS, JavaScript, TypeScript** (**240 respondents**). 9 of the top 10 combinations are built around the HTML/CSS + JavaScript + TypeScript foundation, with C# taking the #10 spot.
 - **Insight:** While exploded data showed JavaScript individually in 1st place, this combination view highlights that modern web development rarely uses JavaScript in isolation—it is overwhelmingly bundled with TypeScript and HTML/CSS as a unified frontend standard.
- **Top 10 Most Common Database Combinations**
 - **Findings:** **PostgreSQL** leads as a standalone choice (**994 respondents**), followed by **Microsoft SQL Server** (**689 respondents**) and **MySQL**. Notably, 6 of the top 10 database slots (positions 4, 5, 7, 8, 9, 10) represent multi-database stacks paired with PostgreSQL, while **MongoDB** holds 6th place.
 - **Insight:** PostgreSQL is not only the single most popular primary database, but it is also the core component around which developers build multi-database hybrid architectures.

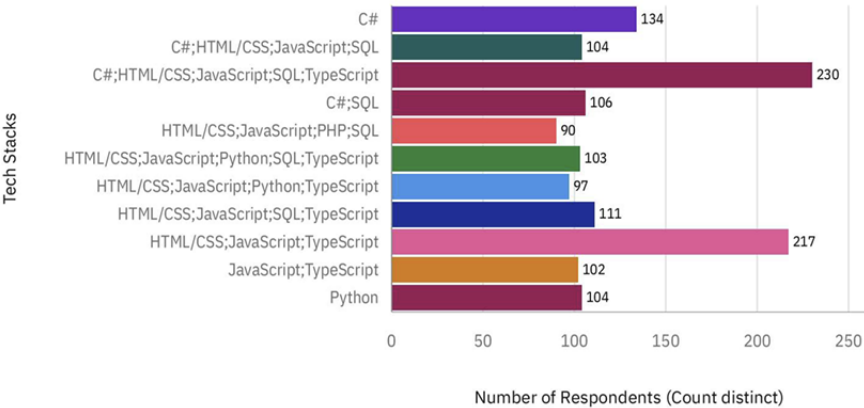
DASHBOARD TAB 1: Current Technology Usage

- **Platform Combinations (Word Cloud)**
 - **Findings:** **AWS** appears as the dominant single keyword, followed by **Azure**. Most other multi-platform combinations feature AWS paired with secondary cloud services.
 - **Insight:** AWS is the primary cloud choice, but multi-cloud strategies are extremely common—most multi-cloud setups pair AWS with either Azure or Google Cloud
- **Top Web Framework Combinations**
 - **Findings:** **Spring Boot** takes the #1 spot, followed by **React** (#2), **Node.js;React** (#3).
 - **Insight:** **Spring Boot** leads enterprise backend framework combinations, while **React** dominates frontend frameworks both as a standalone choice and when paired with **Node.js/Express**

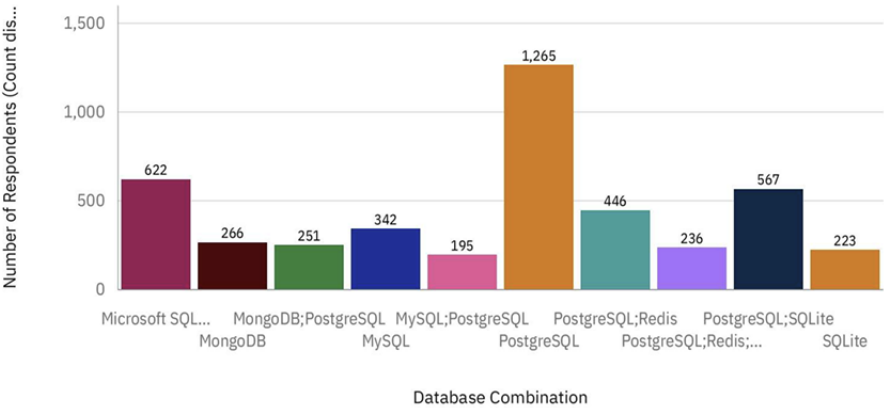
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trend

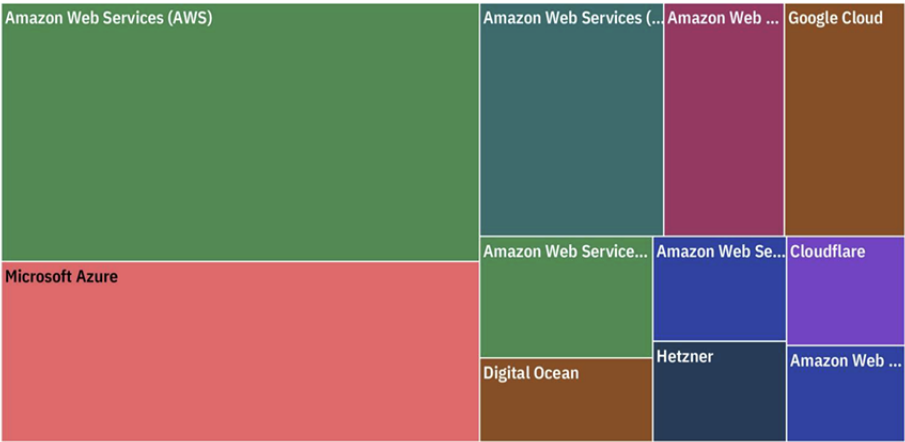
Top 10 Wanted Tech Stacks



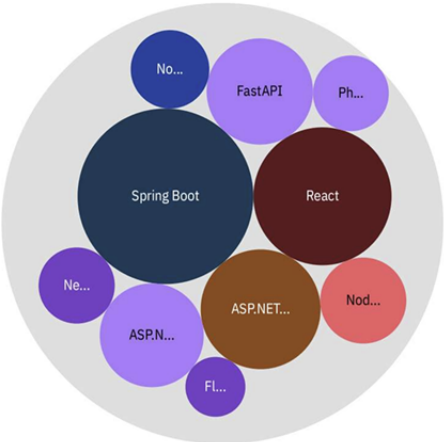
Top 10 Wanted Database Combinations



Top 10 Wanted Platforms Combinations



Top 10 Wanted Webframe Combinations



DASHBOARD TAB 2: Future Technology Trends

- **Top Wanted Tech Stack Combinations (Horizontal Bar Chart)**
 - **Findings:** The top wanted stack is **HTML/CSS; JavaScript; SQL; TypeScript(230)**, followed by **HTML/CSS; JavaScript; TypeScript(217)**. **C#** leads as the top single-language preference (**134**) and Python appears both as a standalone wanted language (**104**) and within full-stack data/web combinations.
 - **Insight:** While **Python** leads as the single most wanted language developers wish to learn, overall demand centers on building full-stack mastery by pairing **HTML/CSS, JavaScript, and TypeScript** with **SQL**. Meanwhile, backend preferences like **C#** and **Python** remain top choices.
- **Top 10 Wanted Database Combinations (Vertical Bar Chart)**
 - **Findings:** **PostgreSQL** as a standalone database leads by a massive margin with **1,265 respondents**. **Microsoft SQL Server** is a distant second with **622 respondents**. Again **PostgreSQL** is in most of the database combinations.
 - **Insight:** Developer demand is heavily concentrating around **PostgreSQL** as the ultimate relational standard—both as a standalone database and combined with others. In the exploded bar chart the clear winner was again **PostgreSQL**.

DASHBOARD TAB 2: Future Technology Trends

- **Top 10 Wanted Platform Combinations (Treemap)**
 - **Findings:** **Amazon Web Services (AWS)** occupies the largest single area, followed by **Microsoft Azure**. **Google Cloud** and **Cloudflare** feature prominently among secondary platforms. Cloud infrastructure alternatives like **Hetzner** and **DigitalOcean** also secure dedicated tiles within developer future preferences.
 - **Insight:** While **AWS** and **Azure** remain the clear cloud priorities for developer learning and adoption, interest in developer-centric edge and hosting solutions like **Cloudflare**, **Hetzner**, and **DigitalOcean** is steadily rising.
- **Top 10 Wanted WebFrame Combinations**
 - **Findings:** **Spring Boot** and **React** remain the two largest desired framework targets. **FastAPI** holds a large standalone circle, followed by **ASP.NET**
 - **Insight:** **Spring Boot** and **React** dominate as the top backend and frontend framework choices developers want to adopt next. Modern API-first and full-stack solutions—specifically **FastAPI**, **ASP.NET Core**, **Node.js**, **Next.js**, **Flask**, and **Phoenix**—round out a diverse landscape focused on performant, highly scalable web services.



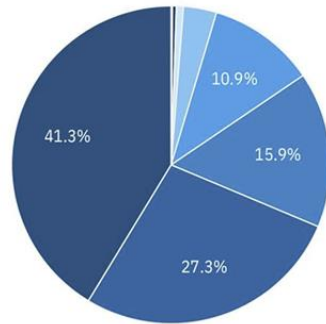
DASHBOARD TAB 3: Demographics

Demographics

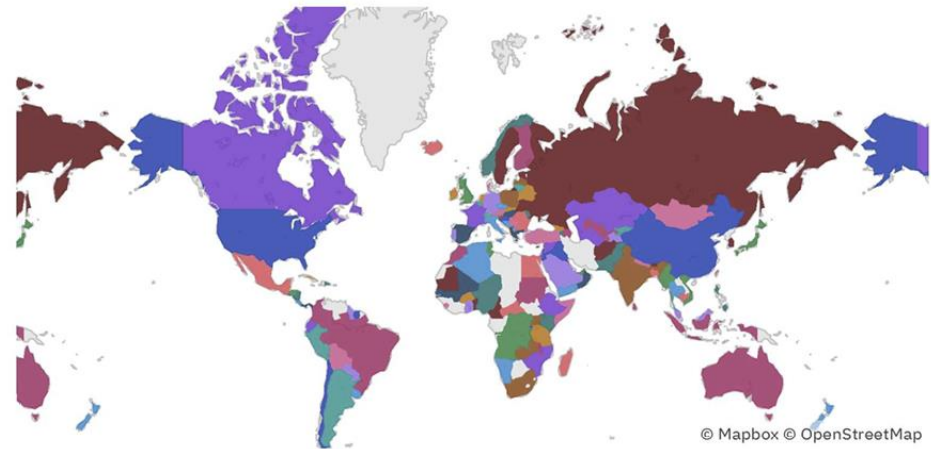
Respondent Distribution by Age

Age

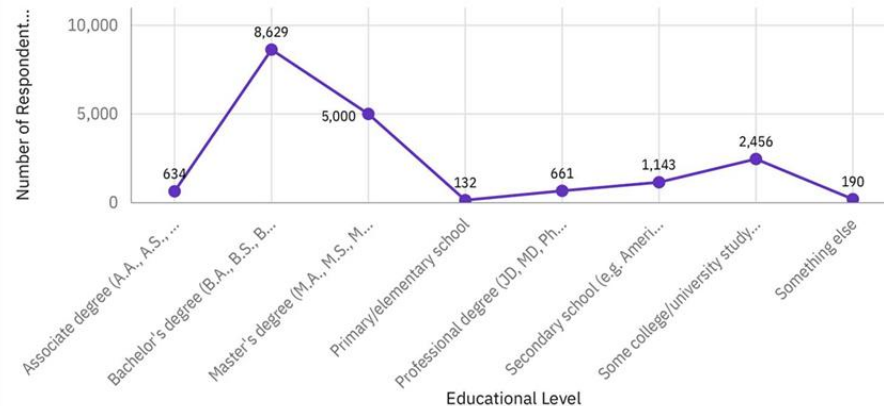
- Prefer not to say
- 65 years or older
- Under 18 years old
- 55-64 years old
- 45-54 years old
- 18-24 years old
- 35-44 years old
- 25-34 years old



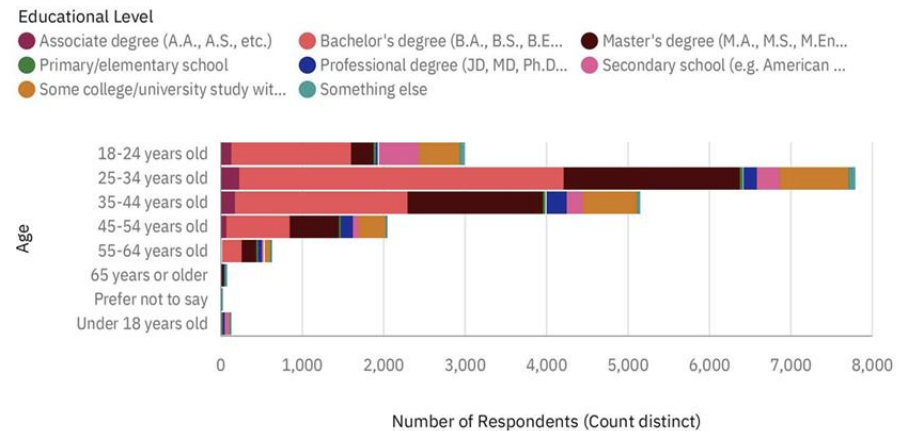
Respondent Distribution by Country



Respondent Distribution by Formal Education Level.



Respondent Distribution by Age and Level of Education



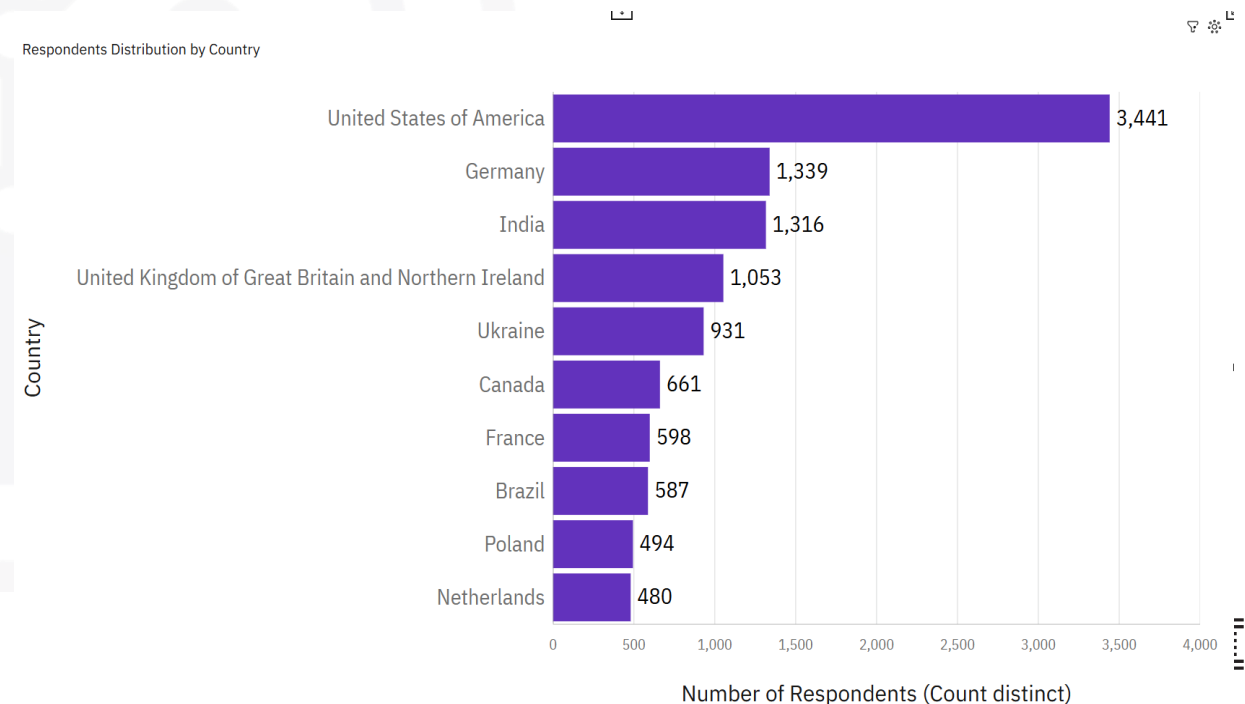
DASHBOARD TAB 3: Demographics

Respondent Distribution by Age (Pie Chart)

- Software development is heavily driven by early-to-mid career professionals, with nearly **70% of respondents** being in the **25-44** age group.

Respondent Distribution by Country (Choropleth Map)

- Software engineering remains a globally dispersed profession with major respondent clusters in key tech hubs across North America, Western Europe, and India.



DASHBOARD TAB 3: Demographics

- **Respondent Distribution by Formal Education Level (Line Chart)**
 - **Bachelor's degree** is by far the most common level of education with **8,629 respondents**. Formal higher education remains the predominant path into software engineering, with **over 70% of respondents holding a Bachelor's or Master's degree**.
- **Respondent Distribution by Age and Level of Education (Stacked Bar Chart)**
 - The vast majority of the respondents have some type of formal education with **Bachelor degree** being the most common. As developers progress into the **35+ age brackets**, the proportion of **Master's degrees** grows significantly relative to **Bachelor's degrees**.

DISCUSSION

- **Current vs. Desired Tech Stack Trends**
- **The Web & Data Baseline:** Stacks combining **HTML/CSS, JavaScript, TypeScript, and SQL** represent the core foundation of active production systems.
- **The Surge Toward Python & Modern Languages:** **Python** leads as the **#1 most desired language** overall (\$12.6\%\$), overtaking JavaScript (\$8.6\%\$) as developers expand into AI and data science workflows. Modern backend standards like **Rust** (\$6.2\%\$) and **Go** (\$5.0\%\$) are climbing rapidly into top future targets.
- **Database & Cloud Consolidation:** **PostgreSQL** is the dominant database choice—leading future interest both standalone and paired with **Redis** for caching. **AWS** remains the primary cloud anchor, flanked by **Azure** and rising edge platforms like **Cloudflare**.



DISCUSSION



- **2. Demographics & Educational Attainment**
- **Core Age Cohort:** The workforce is heavily concentrated in early-to-mid career professionals, with nearly **70% of developers aged 25–44** (led by the 25–34 age group).
- **Formal Education Dominance:** **Bachelor's degrees** represent the most common entryway, while **Master's degrees** become significantly more prevalent in the 35+ age cohort.
- **3. Strategic Recommendations**
- **Recruitment Focus:** Prioritize talent skilled in **TypeScript, Python, and PostgreSQL**, aligning with peak developer interest.
- **Architecture Modernization:** Standardize cloud pipelines around **AWS/Azure** and transition backend systems toward performant frameworks like **Spring Boot, ASP.NET Core, and FastAPI**.





OVERALL FINDINGS & IMPLICATIONS

Key Analytical Findings

- **Market Demand vs. Future Interest (The Tech Stack Pivot):** While production systems rely heavily on web-standard baselines (**HTML/CSS, JavaScript, SQL, C#**), developer upskilling and future interest are pivoting sharply toward **Python, TypeScript**, and high-performance backend tooling (**Rust, Go**).
- **Database Modernization:** **PostgreSQL** has established itself as the unified open-source database standard, leading future adoption both as a primary relational engine and paired with **Redis** for in-memory caching.
- **Cloud & Framework Polarization:** Enterprise infrastructure continues to solidify around **AWS** and **Microsoft Azure**, while web frameworks show a split between established enterprise backends (**Spring Boot, ASP.NET Core**) and fast-growing API/frontend frameworks (**React, FastAPI, Next.js**).
- **Demographics & Educational Thresholds:** The global software industry—spanning major technology hubs and global tech centers—is driven by a core demographic aged **25–44** (68.6% of respondents). Higher education remains a near-universal baseline, with **Bachelor's** and **Master's** degrees representing over 70% of all developers and reaching **100% formal degree attainment** among senior professionals (aged 45+).



OVERALL FINDINGS & IMPLICATIONS



Industry & Business Implications

- **For Talent Acquisition & Hiring:** Companies seeking high retention should hire for current core competencies (**JavaScript/SQL**) while providing clear pathways for engineers to work with high-demand future tech (**TypeScript, Python, PostgreSQL**).
- **For Systems Architecture & IT Strategy:** Engineering teams should standardize relational data layers on **PostgreSQL** to match industry momentum, while designing cloud pipelines to support multi-cloud deployments (**AWS/Azure**) and edge hosting (**Cloudflare, DigitalOcean**).
- **For Developer Training & Upskilling:** L&D programs should prioritize **TypeScript, FastAPI**, and cloud-native architecture to bridge the gap between traditional enterprise development and modern full-stack workflows.



CONCLUSION



- **Production Foundations:** Active workflows remain heavily grounded in established standards—led by **HTML/CSS, JavaScript, SQL**, and enterprise cloud anchors like **AWS**.
- **Future Tech Shifts:** Developer interest is rapidly shifting toward **Python, TypeScript, Rust**, and modern data stacks built around **PostgreSQL** and **Redis**.
- **Workforce Profile:** The industry is anchored by early-to-mid career professionals (aged **25–44**), with formal higher education (**Bachelor's** and **Master's** degrees) acting as a near-universal requirement for career progression
- **Final Recommendation**
 - Align organizational hiring and training with high-demand emerging technologies while maintaining robust infrastructure built on proven industry standards.



APPENDIX

Included in the appendix are supplementary EDA plots supporting our core analytical findings:

1. Technology Demand & Stack Shifts

- Programming Language Usage vs. Future Developer Demand (%)
- Web Framework Preferences mapped against Job Satisfaction Ratings
- Top Preferred Database and Language Distributions across Developer Age Groups

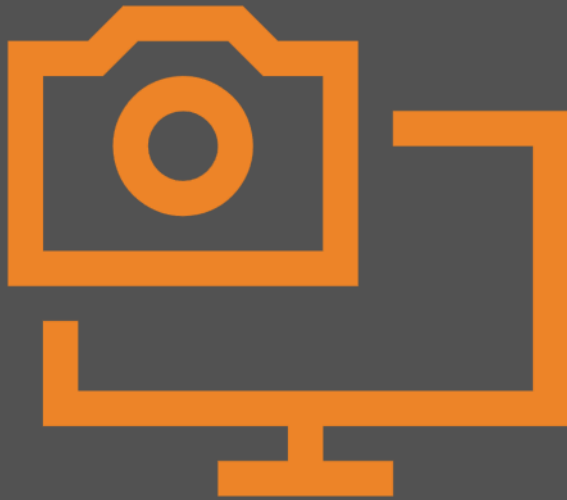
2. Workforce Structure & Career Dynamics

- Employment Type Distributions across Educational Attainment Tiers
- Global Compensation Distributions across UN Geographic Regions (USD, Log Scale)

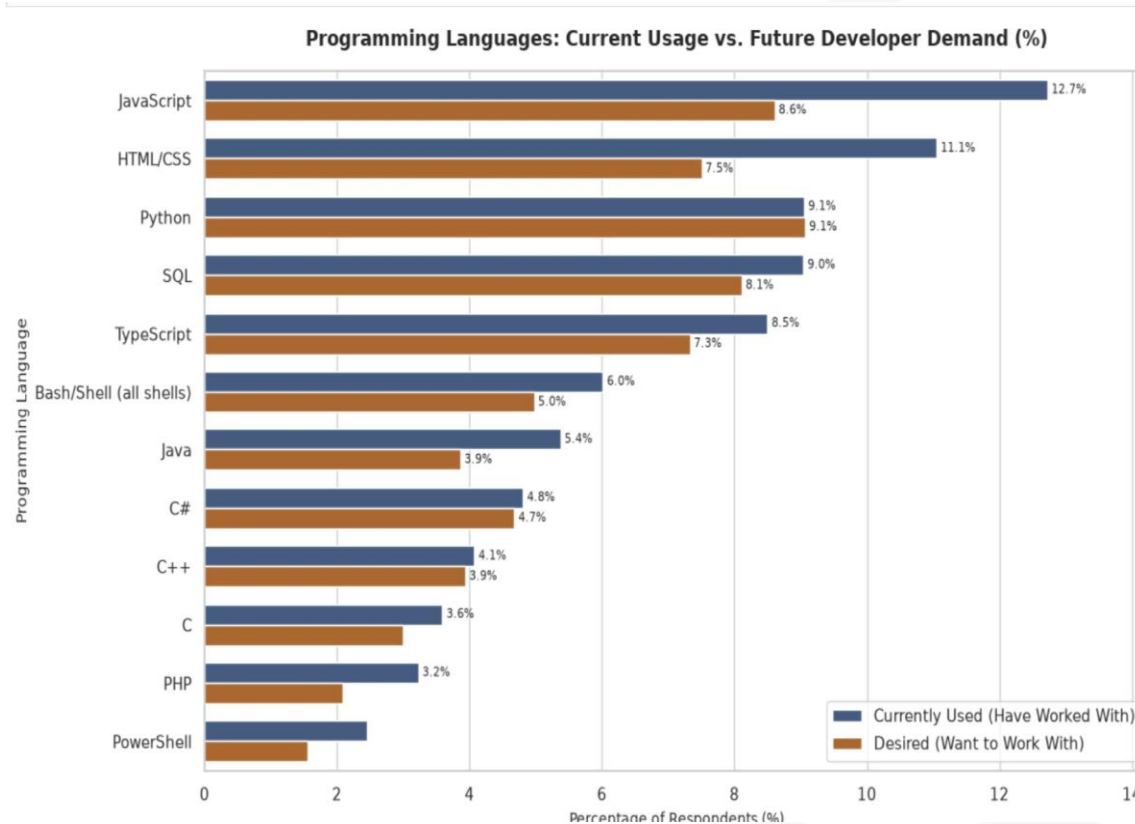
3. Productivity & Developer Tooling

- Top AI Tool Use Cases in Active Software Workflows
- Primary IDE and Collaboration Tool Distributions across Age Cohorts

All notebooks can be found at [Github](#).

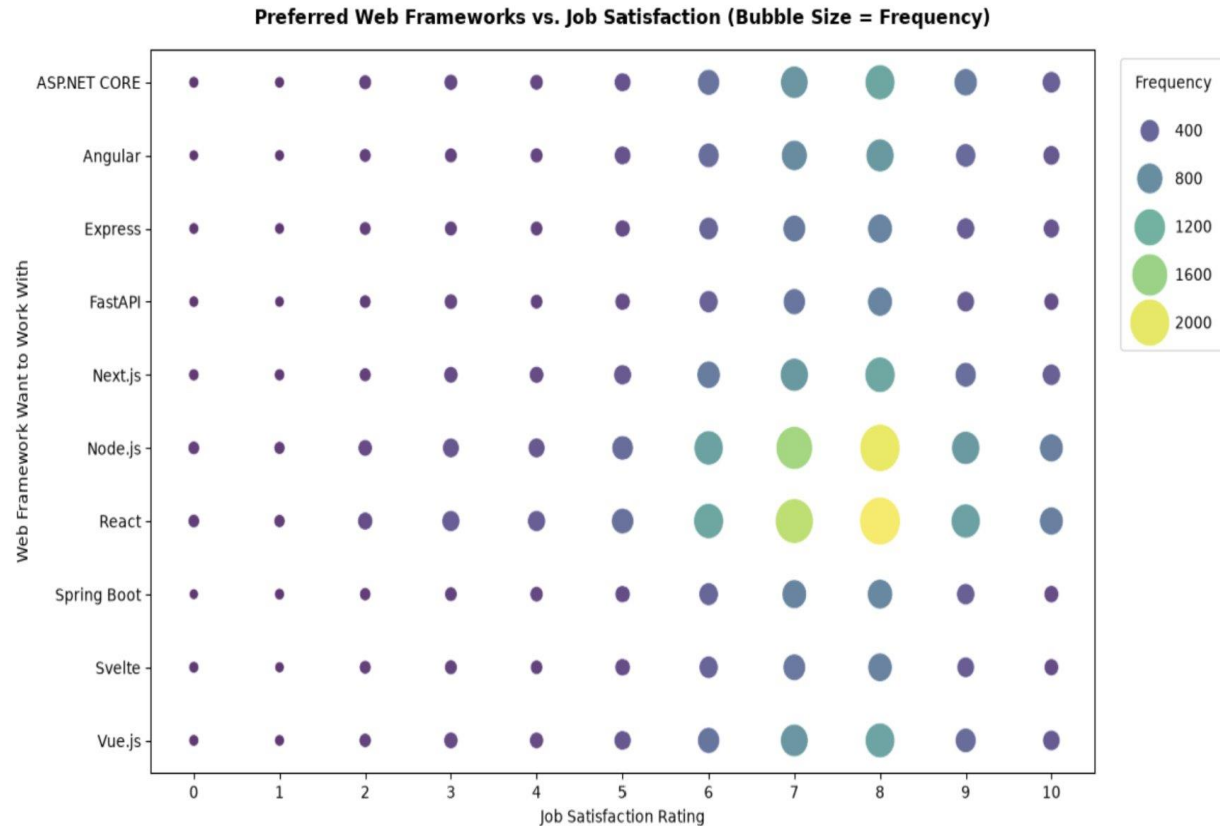


APPENDIX



- While **JavaScript** and **HTML/CSS** hold the highest current production usage, **Python** demonstrates the strongest net retention/growth trajectory. **TypeScript**, **SQL**, **C** and **C#** maintain stable demand, whereas traditional web/shell languages show declining relative growth interest.

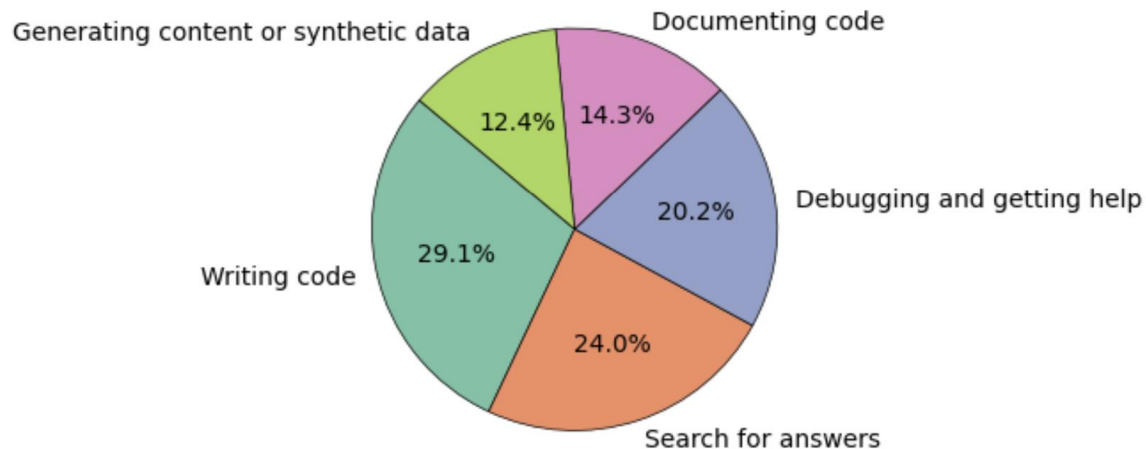
APPENDIX



- Job satisfaction peaks strongly at ratings **7 and 8** across all web frameworks, with **React** and **Node.js** displaying the highest concentration of satisfied developers. Modern frameworks like **FastAPI**, **Next.js**, and **ASP.NET Core** also align with top satisfaction tiers

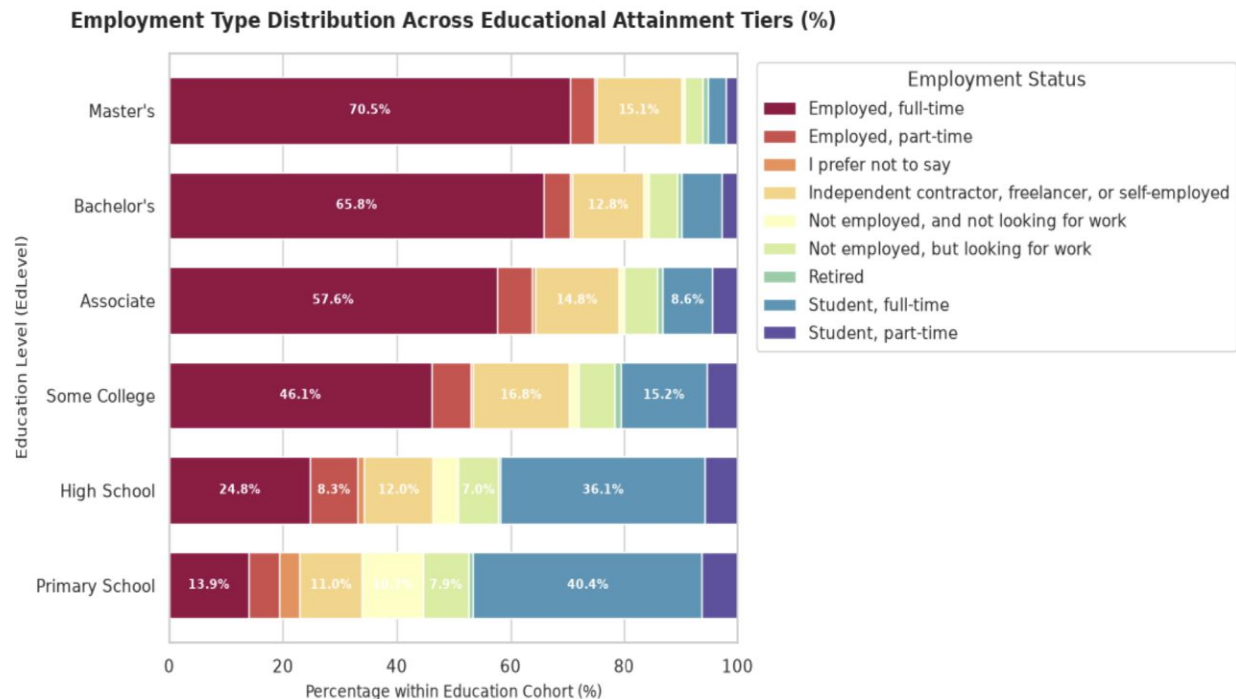
APPENDIX

Top 5 AI Tools Currently Used by Developers



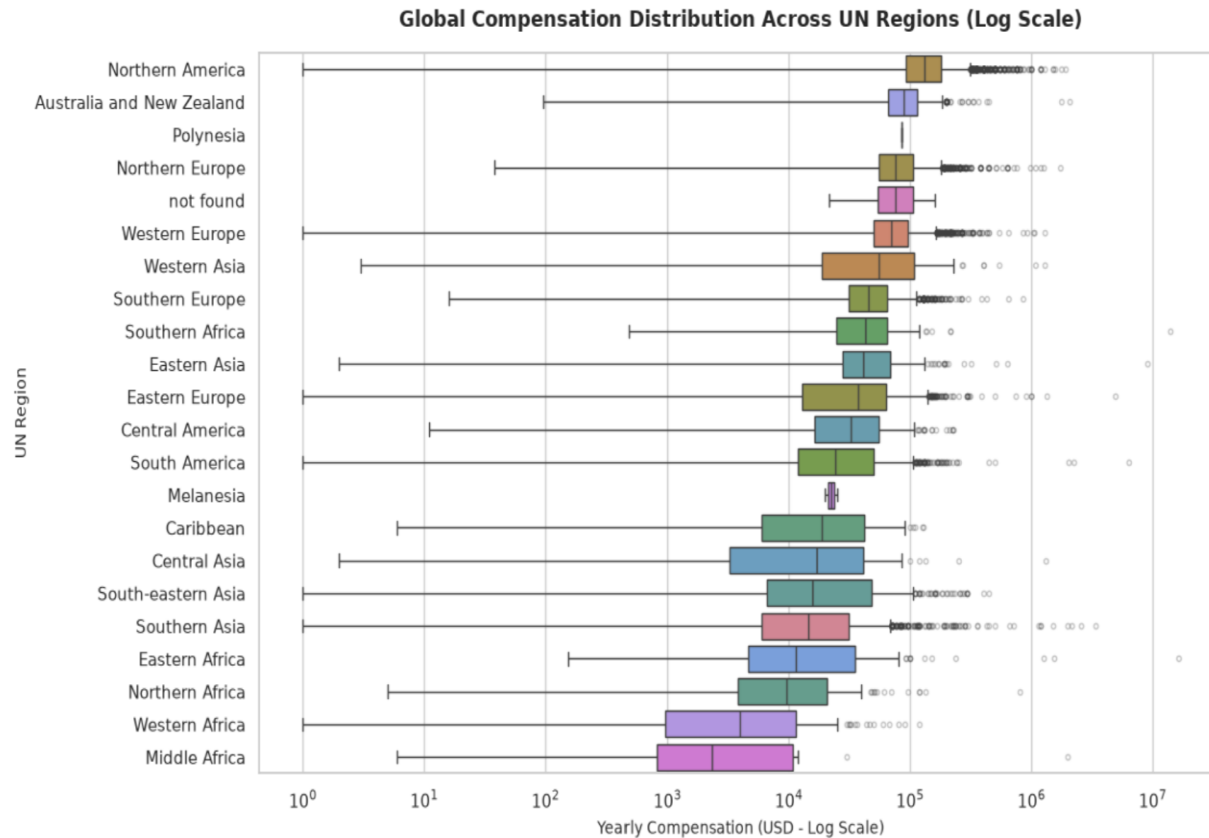
- AI tools are primarily utilized for active development tasks—led by **Writing code (29.1%)** and **Searching for answers (24.0%)**. **Debugging (20.2%)** and **Documenting code (14.3%)** represent major operational productivity use cases.

APPENDIX



- Full-time employment scales directly with formal education, reaching **70.5%** among **Master's** degree holders and **65.8%** among **Bachelor's** degree holders. In contrast, non-degree tiers (**High School, Primary School**) consist predominantly of full-time students (**36.1%** and **40.4%** respectively), confirming that lower formal degree counts in developer surveys are heavily driven by younger individuals actively pursuing education

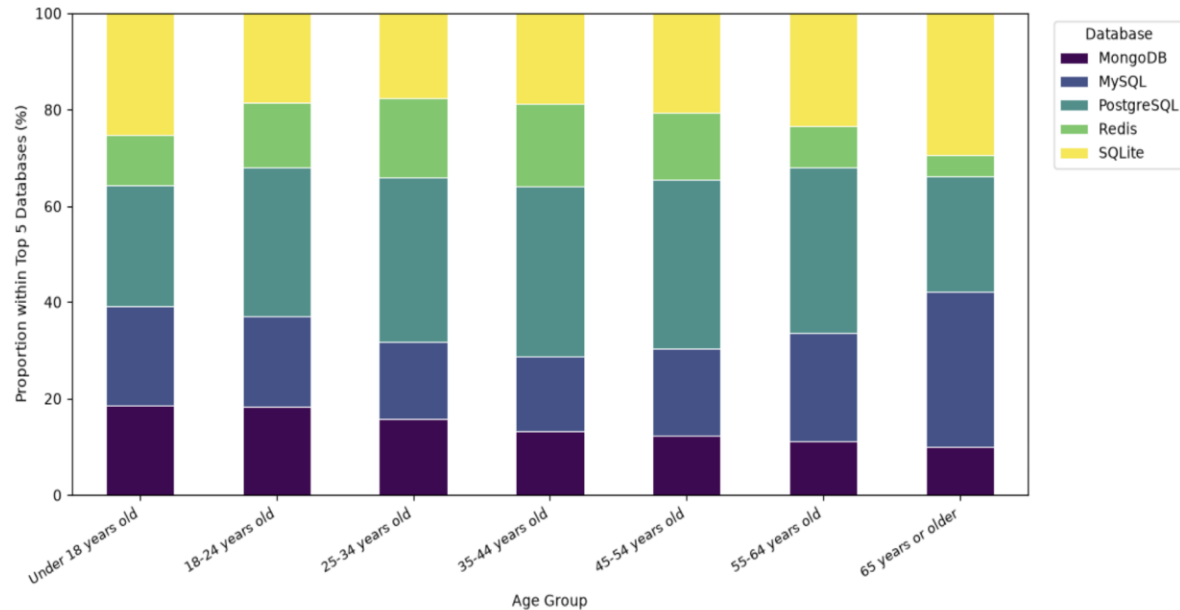
APPENDIX



- Compensation exhibits extreme regional variance globally. **Northern America** leads with the highest median salary ($\sim \$100\text{k} + \$$), followed by **Australia/New Zealand**, **Northern Europe**, and **Western Europe**. Regions in South/Eastern Asia, Eastern Europe, and Africa show lower overall median baselines but maintain significant upper-quantile outliers.

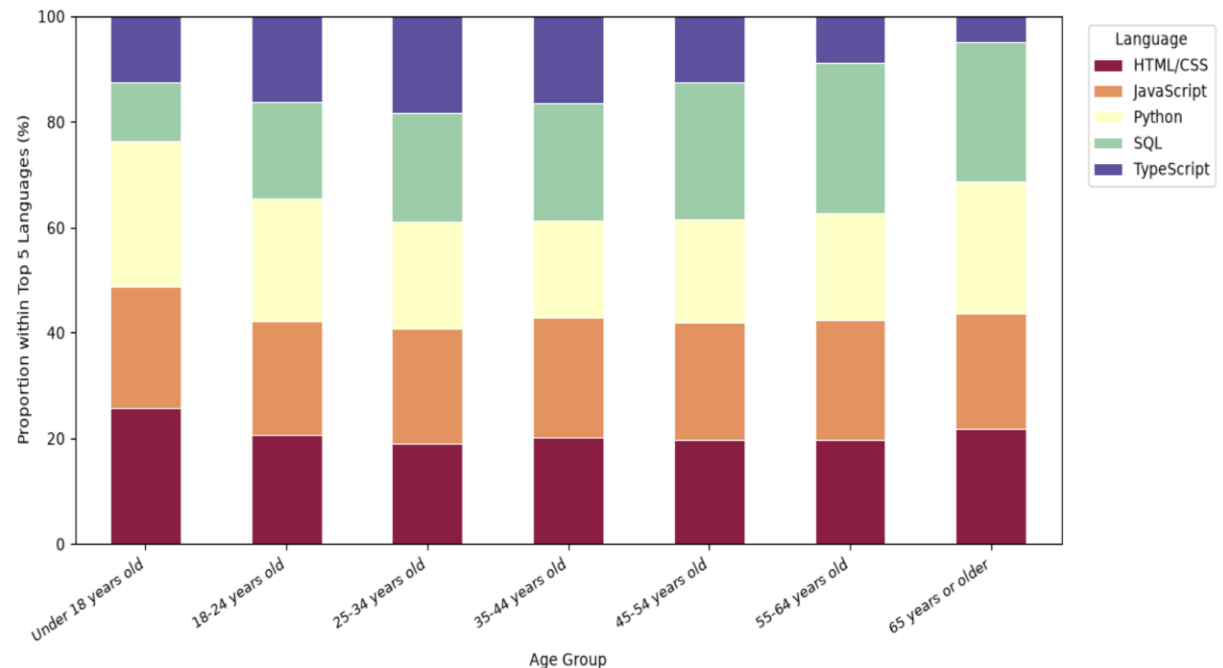
APPENDIX

Proportion of Top 5 Preferred Databases Wanted by Age Group



- **PostgreSQL** (teal) dominates as the most desired database across all working age cohorts)

Proportion of Top 5 Preferred Programming Languages by Age Group



- Language preferences remain consistent across age cohorts